ABDELRAHMAN MAHMOUD

AEROSPACE ENGINEER

PROFILE

Aerospace Engineer, active in the startup ecosystem since 2024. Co-founder of **NestGo**, an EdTech platform that automates skill-gap analysis and curates personalized learning, and **Filmoria**, a Cinematech safe-content hub.
Experienced in managing the complete machine learning lifecycle, with deep expertise in Retrieval-Augmented Generation (RAG) frameworks and LLM orchestration. Supported by comprehensive training at **DEPI**. Backed by industry certifications (Microsoft **AI-900**, **DP-600**, Google Generative AI Leaders **GAIL**, Google Data Analytics Professional)

My focus is translating complex business needs into scalable, data-driven software solutions.

WORK EXPERIENCE

Business Developer, AydnLabs

SEP 2026 - **Now**

- Grow existing business areas and drive sales growth with presentations, proposals, and marketing materials.
- Conduct market research, collect and analyze client data, and identify business opportunities and pain points.
- Prepare regular reports on business development activities, achievements, and challenges.
- Gather feedback from prospects and clients to help improve products, services, and the sales process.

Embedded Systems Engineer, AydnLabs

JUN 2026 - **Now**

- Develop and maintain software code based on defined component requirements and component design.
- Configure the different software components according to the specified requirements.
- Deliver and integrate the software component into the software according to the defined high-level design.
- Develop component test plans and test cases to verify the correct functionality against the requirements.
- Verify the software component functionalities in a defined HW target/platform.
- Report findings (SW Bugs) on the configuration management tool

Co-founder & Technical officer, NestGo

OCT 2025 - APR 2026

- Architected an end-to-end EdTech platform, managing the software lifecycle from system design to deployment.
- Developed automated skill-gap analysis algorithms to compare user profiles against optimal candidate metrics.
- Integrated customized educational content delivery systems based on user assessment data.

Space mission concepts Engineer, AQMAAR.SPACE

JUL 2025 - JAN 2026

- Translated complex aerospace system requirements into production-ready software solutions.
- Directed mission testing teams, established validation protocols, and analyzed system telemetry and data to ensure continuous hardware reliability and successful integration before deployment.
- Conducted comprehensive operational and technical analyses for space missions (earth observation and IoT satellite constellations) to evaluate viability and define mission scope.
- Simulated mission profiles using **MATLAB** to optimize orbital performance, and architected ground station systems to support satellite communications.
- Evaluated **space debris** risks and developed safe disposal methodologies, ensuring end-of-life satellite operations comply with international regulatory frameworks.

F R E E L A N C I N G

Web Developer, Rakaiz AL-Tamdidat JAN 2026

Engineered & deployed the official web platform using React, Node.js, established the company's digital architecture.

French AI Coding Trainer, Invisible Technologies Inc MAY - OCT 2025

Assessed AI-generated responses in STEM topics (Mathematics, Statistics) for scientific accuracy and logical alignment.

AI Coding Evaluator, Scale labs SEP - OCT 2024

- Training generative AI models by evaluating and improving code generated by these systems.
- Crafting and answering questions related to computer science concepts
- Reviewing AI-generated solutions to ensure accuracy and adherence to best practices.

I N T E R N S H I P S

Space Systems Engineer, Egyptian space agency JAN-FEB 2024

- Designing satellite **electrical power systems**, developing energy management algorithms, and conducting performance testing for subsystem switching and power flow.
- **Simulating** space environment hazards to evaluate their effects on satellite subsystems, and executing debris modeling and mitigation strategies.
- Analyzing solar events, cycle indices, and geomagnetic/ionospheric data to **predict** and **forecast** space weather hazards and their impact on satellite signal integrity.
- Engaged in space technology operations, including **CanSat** launching and on-the-job mission simulations using the **Space Keys** platform.

IT Networks engineer, Cairo International Airport JUL-AUG 2023

- Assisted in configuring, patching, and troubleshooting the airport's enterprise LAN/WAN infrastructure and **Cisco** routing/switching hardware.
- Utilized **Kali Linux** for network vulnerability diagnostics and managed Cisco firewall and **VPN** rules for secure vendor connectivity.
- Executed physical infrastructure deployments in primary **data centers**, including server installation, rack assembly, and structured cabling.

Embedded Systems Engineer, National Research Institute of Astronomy and Geophysics AUG-SEP 2023

- Optimized hardware configurations for Field-Programmable Gate Array (**FPGA**) architectures and Integrated Circuit (IC) designs, focusing on embedded system programming.
- Analyzed **satellite communication systems** and aerospace data transmission frameworks, mapping out signal routing and network architectures.
- Evaluated real-world AI case studies, exploring the practical deployment of **machine learning** models to automate seismic data processing and subsurface anomaly detection.

Systems engineer, Egyptian Petrochemicals Holding Company JUL-AUG 2023

- Analyzed the petrochemical value chain, mapping the complex manufacturing progression from raw feedstock selection to the production of primary, intermediate, and downstream end-products.
- Conducted on-site technical evaluations at the Egyptian Petrochemical Company, analyzing Distributed Control Systems (DCS), control elements, and operational trend data across major production units (PVC, VCM, and Chlorine-Caustic Soda).
- Evaluated the broader petrochemical sector landscape, assessing structural challenges, future technological trends, and environmental impact mitigation strategies.

E D U C A T I O N

Bachelor of Navigation science and space technology

Beni Suef University **Sep 2020- July 2025**

- Specialization in **AI** in Space Navigation **CGPA**: 3.69/4
- Advanced **mathematics** | **Probability** & **Statistics** | Mechanical design | **Robotics** | Advanced control | Electronics | Electromagnetics | **Embedded** Systems | Structural Analysis | Product lifecycle | Project management.
- Capstone on "Enhancing Space Mission Planning Using Retrieval-Augmented Generation Models"

Diploma in AI and data science **April- Nov 2024**

Digital Egyptian pioneers initiative (**DEPI**)

- Machine learning algorithms | Attention-based NLP | MLOps with MLflow | Software development
- Azure AI services | GANs training | Prompt Engineering | Supervised & unsupervised ML | Deep learning

PROJECTS

- **Smart Gate Controller**: Refactored legacy embedded firmware into a fully asynchronous architecture to synchronize RFID, QR scanners, and dual-LCD displays with active-low relay physical safety mechanisms.
  Embedded **C/C++** | Hardware-Software Integration | **Microcontroller** Architecture | System Optimization
- **DeepOrbit** (deeporbit.space): Deployed a scalable web platform using React, focusing on component architecture and seamless integration with backend data APIs to ensure high availability.
  React | API Integration | Component **Architecture** | Web Development | **Data-driven** decision-making
- **SIRB Mobile Application** (sirb.space): Built and maintained a cross-platform mobile application using React Native, defining the core software architecture and state management systems to process, route, and display complex system data efficiently.
  React Native | State Management |Data Analysis | Cross-Platform Architecture | **Mobile Development**
- **Autonomous Mission Design**: Led a team to develop a RAG framework that automates CubeSat mission analysis with AI-driven architecture, achieving **a 0.84%** error margin and a user-dependent knowledge base up to 3 GB.
  RAG | Fine-tuning | Prompt Engineering | Langchain | Transformers | PyTorch | Embedding | Chatbot
- **Ionospheric Forecasting**: Engineered a deep learning time-series forecasting model using an LSTM/GRU stack with attention mechanisms to predict VTEC anomalies with **92 %** accuracy on **31 GB** of 3-year row weather data.
  Time-series analysis | **Deep learning** | Large Dataset analysis | TensorFlow

CERTIFICATIONS

- **TOEFL** iBT (Score: **90/120**): An English language proficiency credential validating a CEFR **B2 level**, demonstrating advanced communication skills with section scores in Listening (5.5/6), Speaking (5.0/6), Reading (4.0/6), and Writing (4.0/6).
- Azure AI Fundamentals (**AI-900**): An entry-level certification designed to demonstrate foundational knowledge of Artificial Intelligence (AI) and Machine Learning (ML) concepts on the Microsoft Azure platform.
- Fabric Analytics Engineer (**DP-600**): A certification that validates expertise in implementing analytics solutions using Microsoft Fabric, focusing on data ingestion, transformation, modeling, and visualization.
- Google Cloud Generative AI Leader (**GAIL**): a credential designed for business and tech leaders to validate their ability to harness generative AI for organizational transformation strategically.
- **Google Data Analytics Professional Certificate**: an entry-level credential designed to prepare learners for junior or associate data analyst roles, teaching skills in SQL, Python, spreadsheets, and Tableau through 9 courses

L A N G U A G E S

- **Arabic (Native or Bilingual)**
- **English (Full Professional)**
- **German (Intermediate)**
- **French (Elementary)**